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A Prompt Example

This appendix presents examples of the initial prompts used to guide the penetration tester through various stages of the cybersecurity training process. The two main types of prompts are the **Plan Session** prompt and the **Task Session** prompt.

A.1 Plan Session Initial Prompt

The Plan Session Initial Prompt is used to initiate a structured session, where the assistant helps guide the penetration tester through specific phases of the cybersecurity training process. This prompt defines the overall target and goal for the current phase while referencing the context of the previous phases.

Plan Session Initialization

You are a {name} Assistant running on Kali Linux 2023. Your role is to assist testers in the cybersecurity training process. It includes three stages: reconnaissance, vulnerability scanning, and exploitation. You focus on the {name} phase. You must consider the context of the previous phases if applicable.

Overall Target:

{init_description}

Phase Goal:

{goal}

Optional Reference Tools (may or may not be used):

{tools}

The context of the previous phases:

{context}

Explanation of Variables:

- {name}: Refers to the specific assistant type (e.g., Reconnaissance Assistant, Exploitation Assistant, etc.).
- {init_description}: A brief description of the overall target of the penetration testing process (e.g., *I want to test <target IP>*).
- {goal}: Specifies the specific goal of the current phase (e.g., identify open ports in reconnaissance or exploit a vulnerability in exploitation).
- {tools}: Any reference tools that may assist in achieving the phase goal (e.g., Nmap for reconnaissance, Metasploit for exploitation).
- {context}: The results or insights gained from the previous phases, which may guide the approach for the current phase.

Note that {name}, {goal}, and {tools} are fixed attributes of the assistant character, while {context} is derived from the summarizer's output based on previous phases.

A.2 Task Session Initial Prompt

The Task Session Initial Prompt defines the assistant's behavior in managing individual tasks during the cybersecurity training process. It clarifies how tasks are described in detail and how task results should be evaluated.

Task Session Initialization

You are a {name} Assistant running on Kali Linux 2023. Your role is to assist testers in the cybersecurity training process.

You will receive two types of input:

1. **New Task:** When you receive a New Task, break it down into clear, actionable steps for the tester to follow.
2. **Task Result:** When you receive a Task Result, verify if the task was successful based on the provided result.

Reply with "yes" if you understood.